

Results

01 · RANDOM EFFECTS

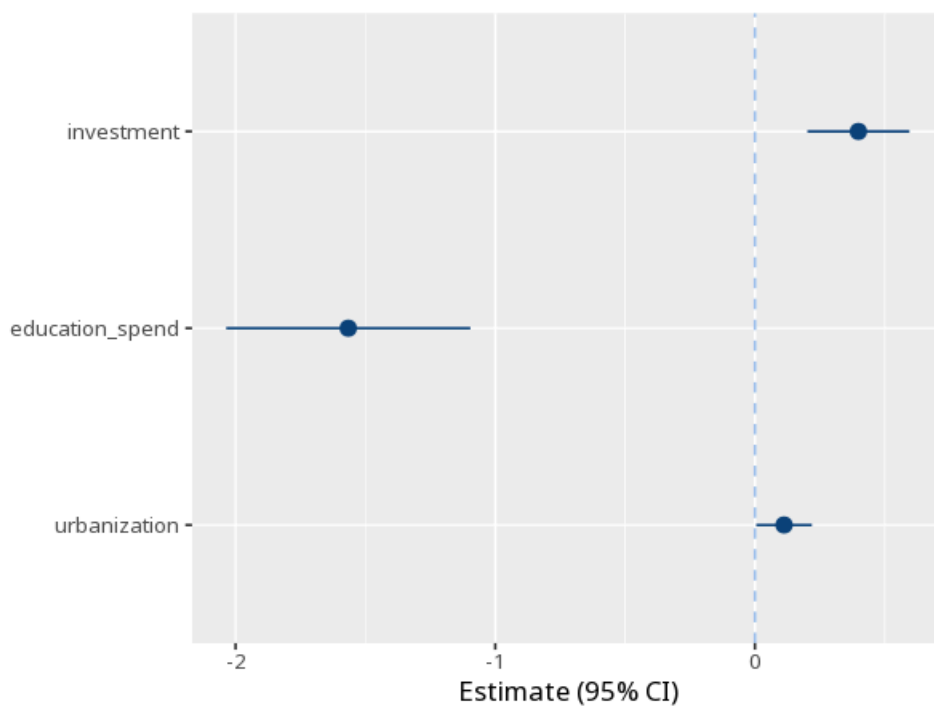
panel regression, random entity effects

Table 1. Coefficients

	(1)
(Intercept)	0.13 (0.47) [−0.79, 1.06]
investment	0.40 (0.10) [0.20, 0.59]
education_spend	−1.57 (0.24) [−2.04, −1.10]
urbanization	0.11 (0.05) [0.00, 0.22]
Num.Obs.	100
N entities	10
R ²	0.93
R ² Adj.	0.93
Within R ²	.36
Between R ²	1.00
Overall R ²	.93

standard errors are clustered by entity. unlike fixed effects, time-invariant predictors can be retained. summary(plm) returns a single R² / adj. R² above; within/between/overall R² (Stata xtreg convention) are also reported below. Breusch–Pagan LM test (RE vs pooled OLS): $\chi^2(1) = 0.73$, $p = .394$ - a low p favours the random effects over pooled OLS. Swamy–Arora $\theta = 0.00$ (variance components: idiosyncratic 0.03, between 0.00).

Figure. Coefficients



Understanding the numbers

B. The estimated effect of a predictor, treating entity differences as random rather than fixed - only trustworthy if a Hausman test favors random effects. Here, $B = 0.40$.

R². plm's own overall model R² for the random-effects fit. Here, $R^2 = 0.93$.

R² Adj.. That R², adjusted for the number of predictors. Here, adjusted $R^2 = 0.93$.

Within R². How well the predictors explain variation WITHIN each entity over time (Stata xtreg convention). Here, within $R^2 = .36$.

Between R². How well the predictors explain differences in the ENTITY-level averages. Here, between $R^2 = 1.00$.

Overall R². How well the predictors explain the outcome using the raw, untransformed data. Here, overall $R^2 = .93$.

N. The total number of entity-period observations used to fit the model. Here, $N = 100$.

N entities. The number of distinct entities in the panel. Here, $N \text{ entities} = 10$.

Breusch-Pagan LM. A test of random effects against pooled OLS - a low p favours random effects. Here, the Breusch-Pagan LM test gave $\chi^2(1) = 0.73$, $p = .394$.

How to read this test

Like fixed effects but treats entity differences as random, allowing time-invariant predictors. Read each B/p as usual - but only trust this model if the Hausman test favours random over fixed effects. Method: R plm package (Croissant & Millo, 2008). Your significance threshold (α) is 0.05.

APA template: In a random-effects model, predictor investment gave $B=0.40$, $p < .001$.

Statistical basis: Random-effects (GLS) panel estimator (Croissant, Y., Millo, G. (2008). "Panel Data Econometrics in R: The plm Package." Journal of Statistical Software, 27(2), 1-43.). Full references in the exported CITATIONS.txt.

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